

# INSTITUTE OF ENGINEERING AND MANAGEMENT, KOLKATA

Department of Computer Science & Engineering

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## LABORATORY ASSIGNMENT REPORT ASSIGNMENT 4: ASSERTIONS AND VERIFICATION

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|-------------------------|---|-----------------------|
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| <b>Department</b>       | : | CSE (AI & ML)         |
| <b>Course</b>           | : | Python for Automation |
| <b>Academic Session</b> | : | 2023-2027             |
| <b>Submission Date</b>  | : | 28 Sept, 2026         |

## 1. MODULE NAME & EXPERIMENT TITLE

**Module Name:** Assertions and Verification

**Experiment Title:** Implementation of Test Assertions and Verification of REST API Endpoint Responses using "Should Be Equal As Strings" in Robot Framework

## 2. PROBLEM STATEMENT

Develop automated test scripts in Robot Framework to validate test state and API service integration through the following tasks:

- Write a test case that performs an action and then uses an assertion to check whether the expected outcome matches the actual result.
- Implement a test case that verifies the response of an API request using the "Should Be Equal As Strings" keyword.

## 3. OBJECTIVE

- To understand basic and advanced assertion techniques in automated test design.
- To send HTTP requests (GET) to a RESTful web service API using Robot Framework's RequestsLibrary.
- To extract JSON response payload data and validate specific response elements (such as HTTP Status Code and String fields) using verification keywords like Should Be Equal As Strings.

## 4. TOOLS, SOFTWARE, AND CONCEPTS USED

- **Programming Language:** Python 3.11.x
- **Automation Framework:** Robot Framework 6.1
- **External Library:** robotframework-requests (RequestsLibrary for API automation)
- **Target REST API Endpoint:** JSONPlaceholder REST API (<https://jsonplaceholder.typicode.com>)
- **IDE / Development Environment:** Visual Studio Code (VS Code)
- **Core Concepts:** Assertions, Expected vs. Actual Value Comparison, HTTP GET Request, REST API Response Validation, JSON Parsing.

## 5. SOURCE CODE / IMPLEMENTATION STEPS

Test Suite Script (assignment4\_tests.robot)

assignment4\_tests.robot - Robot Framework Test Suite

Robot Framework

```
1  *** Settings ***

2  Documentation Assignment 4 - Assertions and API Verification

3  Library RequestsLibrary

4  Library Collections

5  Library BuiltIn
```

## 6. OUTPUT WITH CLEARLY LABELLED SCREENSHOTS

The execution was carried out using `robot assignment4_tests.robot` in the VS Code terminal. The test engine successfully connected to the online REST endpoint, parsed the response, and evaluated both string and integer assertions.

**Screenshot 1: Terminal Execution Log for Assignment 4 Test Cases**

```
PS C:\Users\Ashmita\Automation_Lab> robot assignment4_tests.robot
===== Assignment4 Tests
===== Computed Sum: 50
TC_01: Basic Action and Assertion Validation | PASS | -----
----- Fetched API Post Title: sunt aut facere repellat provident
occaecati excepturi option... TC_02: Verify API Response String Using Should Be Equal As Strings |
PASS | ===== Assignment4
Tests | PASS | 2 tests, 2 passed, 0 failed
===== Output:
C:\Users\Ashmita\Automation_Lab\output.xml Log: C:\Users\Ashmita\Automation_Lab\log.html Report:
C:\Users\Ashmita\Automation_Lab\report.html PS C:\Users\Ashmita\Automation_Lab>
```

**Screenshot 2: Detailed HTML Test Execution Report (report.html)**

### Robot Framework Test Execution Summary: ALL 2 TESTS PASSED

|                    |                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------|
| Test Suite:        | Assignment4 Tests                                                                                       |
| Target API:        | <a href="https://jsonplaceholder.typicode.com/posts/1">https://jsonplaceholder.typicode.com/posts/1</a> |
| HTTP Status:       | 200 OK (Verified via Should Be Equal As Strings)                                                        |
| Assertions Logged: | Integer comparison (50 == 50), String comparison ('actual_title' == 'EXPECTED_TITLE').                  |

## 7. BRIEF RESULT, OBSERVATION, AND CONCLUSION

### Brief Result:

Both test cases passed successfully. The action-assertion calculation verified that  $15 + 35$  equals 50, and the API GET request successfully retrieved post #1 from JSONPlaceholder and validated its 'title' string field using 'Should Be Equal As Strings'.

### Observation:

- The keyword `Should Be Equal As Strings` converts both arguments to string representation before comparison, which avoids type mismatch errors when comparing numeric HTTP status codes (e.g. integer 200 vs string "200").
- `RequestsLibrary` allows easy integration of REST API testing directly inside standard Robot Framework test scripts without requiring external tools like Postman.
- Dictionaries returned from ``response.json()`` can be queried using ``Get From Dictionary`` keyword provided by standard ``Collections`` library.

### Conclusion:

In this experiment, I successfully implemented test assertions in Robot Framework. I learned how to automate REST API testing by creating HTTP sessions, sending GET requests, parsing JSON response data, and verifying actual API output against expected values using the `Should Be Equal As Strings` assertion keyword.